

Introduction to Network Management



- What is network management?
- Importance in Linux system administration
- Tools & utilities overview (CLI-focused)

/etc/hosts File



- ❑ Maps IP addresses to hostnames on the network.

- ❑ Similar to the hosts file in Windows.

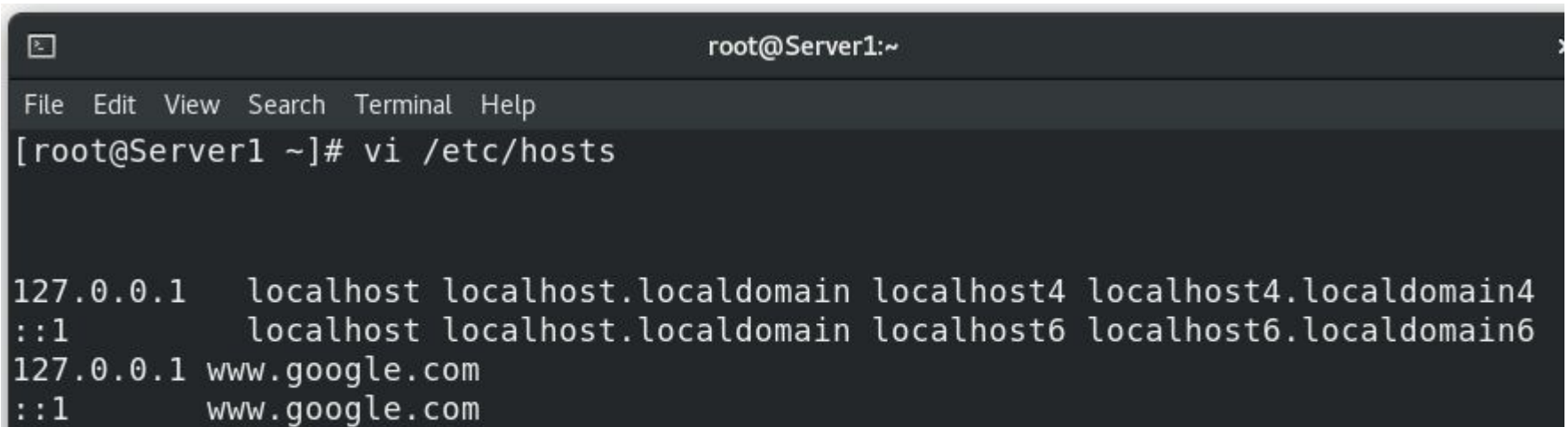
- ❑ File syntax:

`IP_address<Tab>Fully.Qualified.Domain.Name<space>[host_alias]*`

- ❑ Example:

```
192.168.1.10      khtn.edu.vn      khtn
```

- ❑ Applications will consult this file first when resolving a hostname.



```
root@Server1:~  
File Edit View Search Terminal Help  
[root@Server1 ~]# vi /etc/hosts  
  
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4  
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6  
127.0.0.1    www.google.com  
::1         www.google.com
```

Understanding Network Interfaces

- ❑ Types of network interfaces (e.g., `eth0`, `ens33`, `lo`)
- ❑ Check interfaces with:

```
ip link show
```

```
ip link show [network_interface]
```

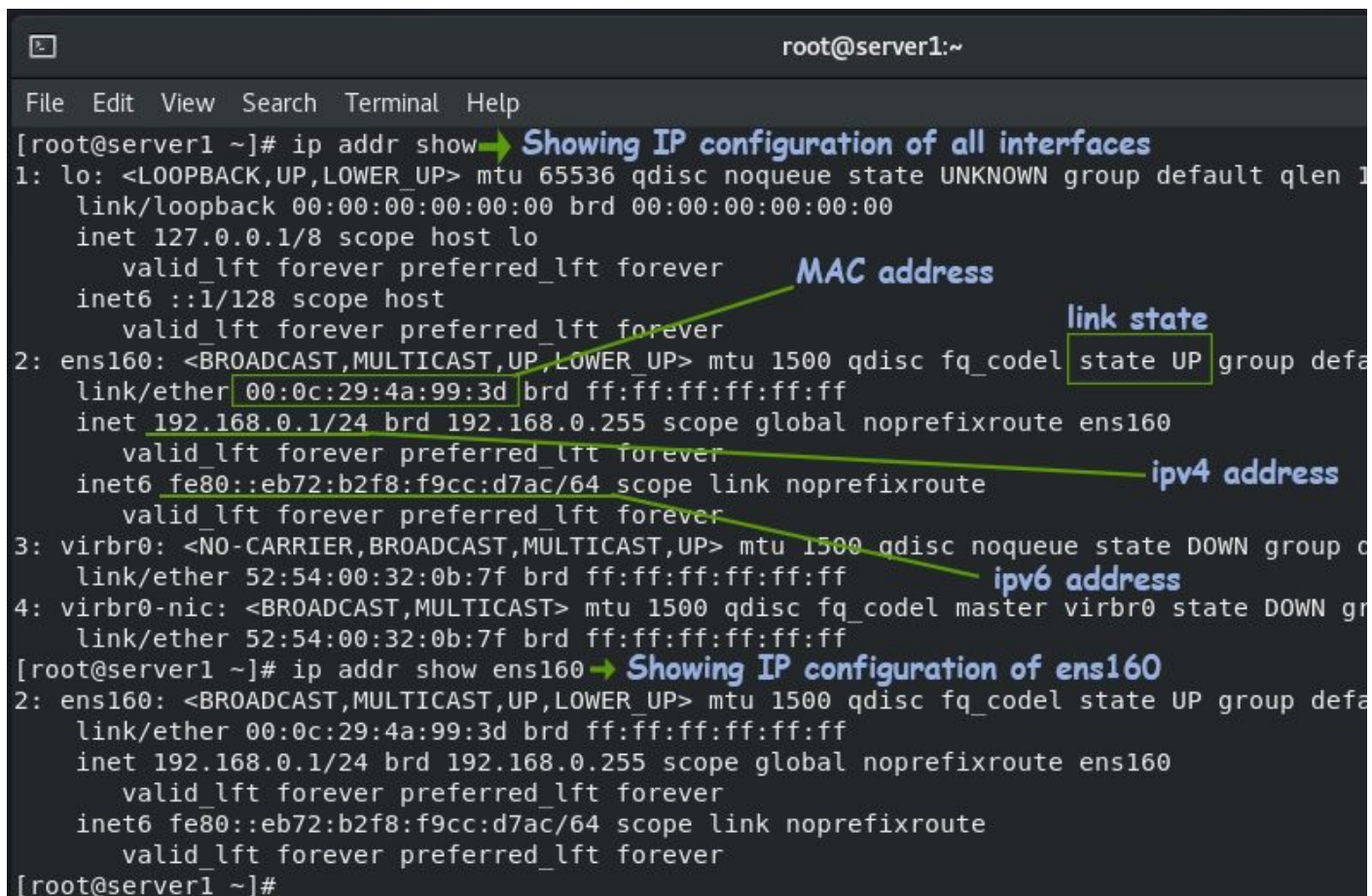
```
root@server1:~  
File Edit View Search Terminal Help  
[root@server1 ~]# ip link show → Showing link state of all interfaces  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT g  
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode  
   link/ether 00:0c:29:4a:99:3d brd ff:ff:ff:ff:ff:ff  
3: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN m  
   link/ether 52:54:00:32:0b:7f brd ff:ff:ff:ff:ff:ff  
4: virbr0-nic: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel mnoqueue state DOWN m  
   link/ether 52:54:00:32:0b:7f brd ff:ff:ff:ff:ff:ff  
[root@server1 ~]# ip link show ens160 → Showing link state of ens160  
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode  
   link/ether 00:0c:29:4a:99:3d brd ff:ff:ff:ff:ff:ff  
[root@server1 ~]#
```

Checking IP Configuration

❑ View IP addresses:

```
ip addr show
```

```
ip addr show [network_interface]
```



The image shows a terminal window with the command `ip addr show` and its output. Annotations highlight specific parts of the output:

- Showing IP configuration of all interfaces**: Points to the command `ip addr show`.
- MAC address**: Points to the MAC address `00:0c:29:4a:99:3d` for interface `ens160`.
- link state**: Points to the state `UP` for interface `ens160`.
- ipv4 address**: Points to the IPv4 address `192.168.0.1/24` for interface `ens160`.
- ipv6 address**: Points to the IPv6 address `fe80::eb72:b2f8:f9cc:d7ac/64` for interface `ens160`.

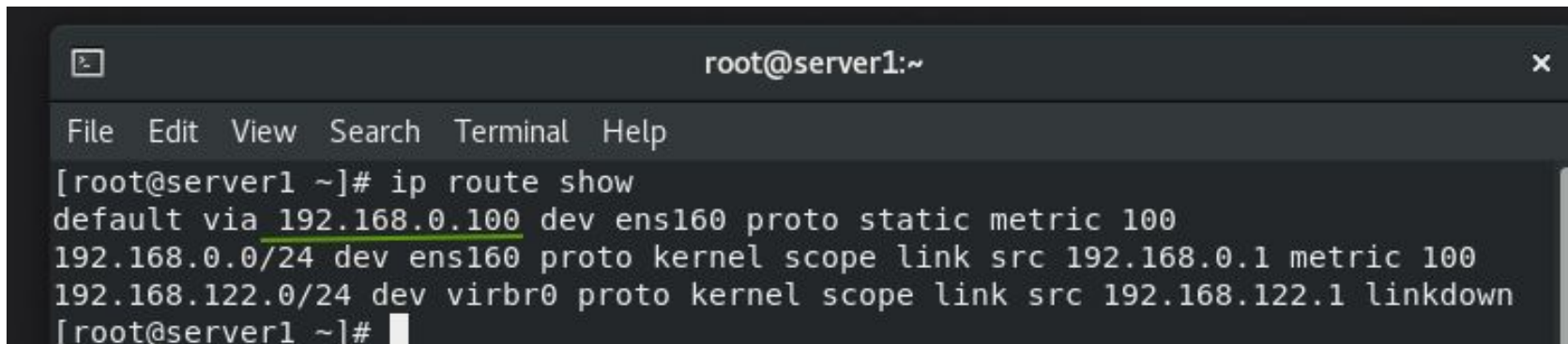
```
root@server1:~  
File Edit View Search Terminal Help  
[root@server1 ~]# ip addr show → Showing IP configuration of all interfaces  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default  
    link/ether 00:0c:29:4a:99:3d brd ff:ff:ff:ff:ff:ff  
    inet 192.168.0.1/24 brd 192.168.0.255 scope global noprefixroute ens160  
        valid_lft forever preferred_lft forever  
    inet6 fe80::eb72:b2f8:f9cc:d7ac/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
3: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default  
    link/ether 52:54:00:32:0b:7f brd ff:ff:ff:ff:ff:ff  
4: virbr0-nic: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel master virbr0 state DOWN group default  
    link/ether 52:54:00:32:0b:7f brd ff:ff:ff:ff:ff:ff  
[root@server1 ~]# ip addr show ens160 → Showing IP configuration of ens160  
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default  
    link/ether 00:0c:29:4a:99:3d brd ff:ff:ff:ff:ff:ff  
    inet 192.168.0.1/24 brd 192.168.0.255 scope global noprefixroute ens160  
        valid_lft forever preferred_lft forever  
    inet6 fe80::eb72:b2f8:f9cc:d7ac/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
[root@server1 ~]#
```

Checking IP Configuration

❑ Show routing table:

```
ip route show
```

This command displays the default gateway IP address. The default gateway connects the host to the remote network.

A terminal window titled 'root@server1:~' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows the command '[root@server1 ~]# ip route show' and its output: 'default via 192.168.0.100 dev ens160 proto static metric 100', '192.168.0.0/24 dev ens160 proto kernel scope link src 192.168.0.1 metric 100', and '192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown'. The prompt '[root@server1 ~]#' is visible at the bottom with a cursor.

```
root@server1:~  
File Edit View Search Terminal Help  
[root@server1 ~]# ip route show  
default via 192.168.0.100 dev ens160 proto static metric 100  
192.168.0.0/24 dev ens160 proto kernel scope link src 192.168.0.1 metric 100  
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown  
[root@server1 ~]#
```

ifconfig Command (deprecated)



- ❑ The **ifconfig** command is deprecated. The **ip** command replaces the **ifconfig** command.
- ❑ In earlier versions of Linux, the **ifconfig** command was the default utility for checking and verifying IP configuration.

ifconfig command	Task	ip command
ifconfig	Show ip configuration of all interfaces	ip address show
ifconfig <i>[interface]</i>	Show ip configuration of the specified interface	ip address <i>[interface]</i>
ifconfig <i>[interface]</i> down	Bring the specified interface down	ip link set <i>[interface]</i> down
ifconfig <i>[interface]</i> up	Bring the specified interface up	ip link set <i>[interface]</i> up
ifconfig <i>[interface]</i> <i>[ip address]</i> ifconfig <i>[interface]</i> <i>[subnet mask]</i>	Assign a temporary IP address to the specified interface	ip address add <i>[ip address/subnet mask]</i> dev <i>[interface]</i>

ifconfig Command vs ip Command

```
net-tools 2.10-alpha
[root@server1 ~]# ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 0.0.0.0 netmask 0.0.0.0 broadcast 255.255.255.255
    ether 00:0c:29:4a:99:3d txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 16 bytes 1392 (1.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 120 bytes 9592 (9.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 120 bytes 9592 (9.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

virbr0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 192.168.122.1 netmask 255.255.255.0 broadcast 192.168.122.255
    ether 52:54:00:32:0b:7f txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@server1 ~]#
```

```
[root@server1 ~]# ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN
    group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_code
    l state UP group default qlen 1000
    link/ether 00:0c:29:4a:99:3d brd ff:ff:ff:ff:ff:ff
    inet 0.0.0.0/0 brd 255.255.255.255 scope global noprefixroute e
    ns160
        valid_lft forever preferred_lft forever
    inet 20.0.0.1/8 scope global ens160
        valid_lft forever preferred_lft forever
3: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noque
    ue state DOWN group default qlen 1000
    link/ether 52:54:00:32:0b:7f brd ff:ff:ff:ff:ff:ff
    inet 192.168.122.1/24 brd 192.168.122.255 scope global virbr0
        valid_lft forever preferred_lft forever
4: virbr0-nic: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel master
    virbr0 state DOWN group default qlen 1000
    link/ether 52:54:00:32:0b:7f brd ff:ff:ff:ff:ff:ff
[root@server1 ~]#
```

```
[root@server1 ~]# ifconfig ens160 down
[root@server1 ~]#
[root@server1 ~]# ifconfig ens160 up
[root@server1 ~]#

[root@server1 ~]# ip link set ens160 down
[root@server1 ~]#
[root@server1 ~]# ip link set ens160 up
[root@server1 ~]#
```

Managing Network Services (systemd & NetworkManager)

NetworkManager: A tool for managing network connections on Linux.

- It can manage wired (Ethernet), wireless (Wi-Fi), VPN connections...
- Main tools: nmcli and nmtui
- Start/enable/disable/restart networking service:

```
systemctl restart NetworkManager
```

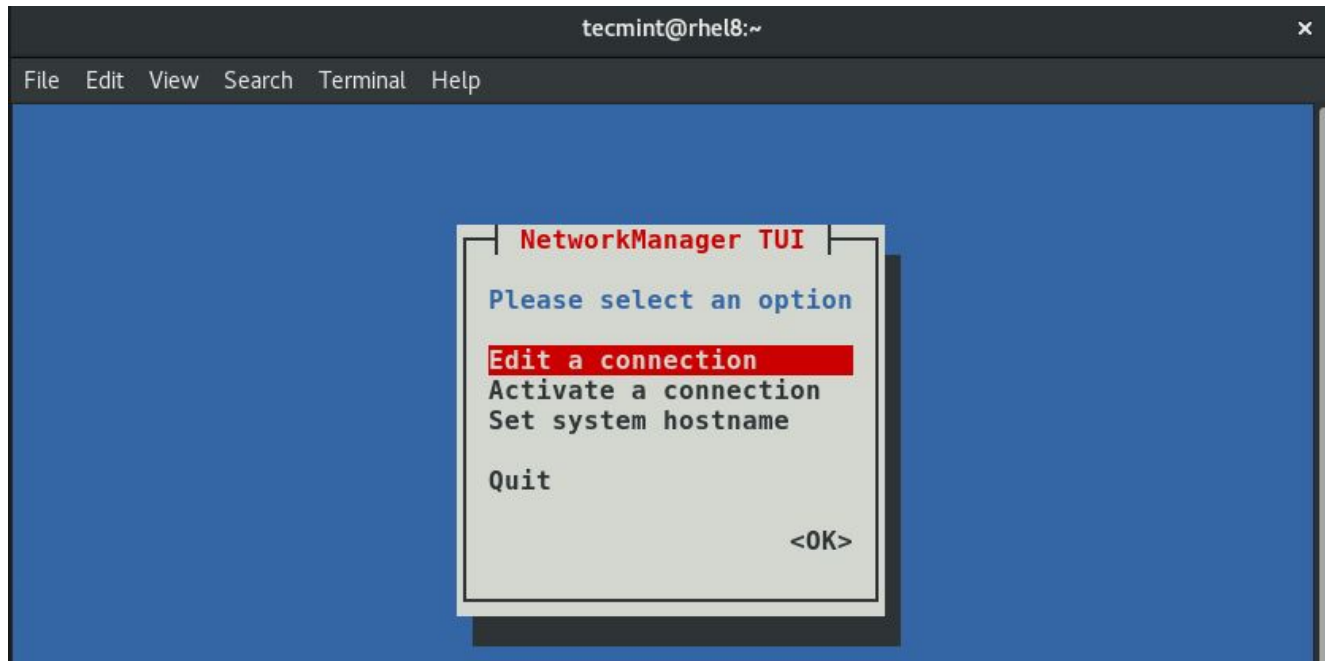
```
systemctl enable NetworkManager
```

```
root@server1:~  
File Edit View Search Terminal Help  
[root@server1 ~]# systemctl status NetworkManager  
● NetworkManager.service - Network Manager  
   Loaded: loaded (/usr/lib/systemd/system/NetworkManager.service; enabled;  
   Active: active (running) since Sun 2022-07-24 19:05:53 IST; 1 months 8 da  
     Docs: man:NetworkManager(8)  
  Main PID: 1164 (NetworkManager)  
    Tasks: 3 (limit: 11076)  
   Memory: 7.6M  
    CGroup: /system.slice/NetworkManager.service  
            └─1164 /usr/sbin/NetworkManager --no-daemon
```

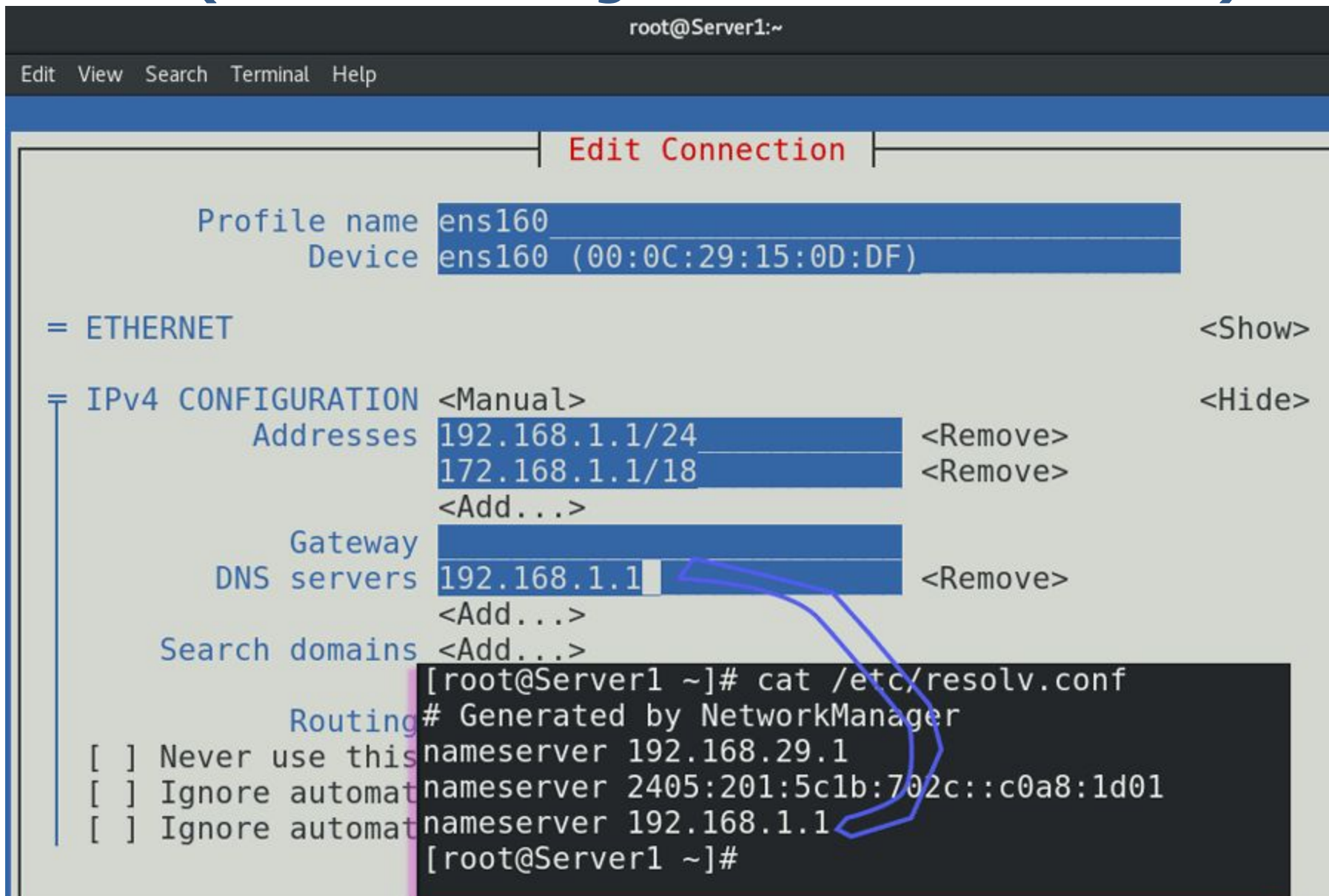
nmtui (NetworkManager Text User Interface)

- nmtui

- Activate a connection: Enable or disable network connections.
- Edit a connection: Create, modify, or delete network connections (Ethernet, Wi-Fi).
- Set system hostname: Configure the system's hostname.



nmtui (NetworkManager Text User Interface)



Assigning Static IP Address (CLI)



- ❑ Files to Edit: `/etc/sysconfig/network-scripts/ifcfg-<interface>`

- ❑ Example:

```
BOOTPROTO=static
```

```
ONBOOT=yes
```

```
IPADDR=192.168.1.100
```

```
NETMASK=255.255.255.0
```

```
GATEWAY=192.168.1.1
```

```
DNS1=8.8.8.8
```

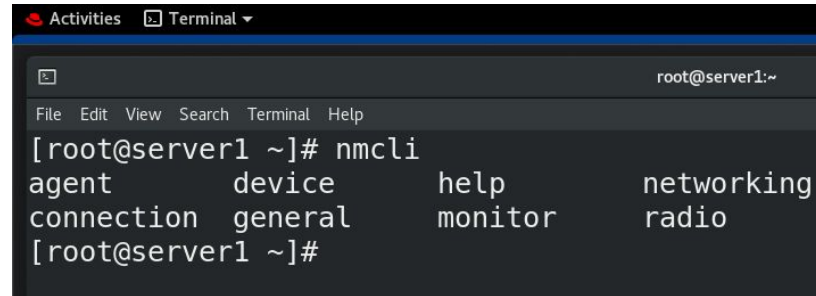
- ❑ Reload settings:

```
# nmcli connection reload
```

```
# systemctl restart NetworkManager
```

nmcli Command

- ❑ Syntax: #nmcli [options] [section] [action]
- ❑ Section:

A terminal window titled 'Terminal' with a menu bar (File, Edit, View, Search, Terminal, Help) and a prompt 'root@server1:~'. The user has entered 'nmcli' and the terminal displays a list of available sections: agent, device, help, networking, connection, general, monitor, and radio.

```
root@server1 ~]# nmcli
agent      device    help      networking
connection general    monitor   radio
root@server1 ~]#
```

Section	Description
help	Used to get help related to the nmcli options and parameters
general	Used to get the status and global configuration of the NetworkManager
networking	Used to start, restart, and manage NetworkManager
radio	Used to manage wireless devices and protocols
connection	Used to manage connections
device	Used to manage network devices
agent	Used to configure and manage various security settings
monitor	Used to monitor network changes

nmcli Command Examples

- ❑ List all available network devices and their current status

nmcli dev status

```
[root@server1 ~]# nmcli dev status
DEVICE          TYPE          STATE          CONNECTION
ens160          ethernet      connected      ens160
virbr0          bridge        connected (externally) virbr0
lo              loopback      unmanaged      --
virbr0-nic      tun           unmanaged      --
[root@server1 ~]#
```

- ❑ A device can have multiple connections, view all of them.

nmcli con show

```
[root@server1 ~]# nmcli con show
NAME          UUID          TYPE          DEVICE
ens160        ed943550-b2fd-4247-89ef-cfc1384162cf  ethernet      ens160
virbr0        f3b19b6b-e1c3-499a-94eb-5db388e56f52  bridge        virbr0
[root@server1 ~]#
```

nmcli Command Examples

- ❑ Lists all configured values of the **ens160** connection

nmcli con show ens160

```
[root@server1 ~]# nmcli con show
```

NAME	UUID	TYPE	DEVICE
ens160	ed943550-b2fd-4247-89ef-cfc1384162cf	ethernet	ens160
virbr0	f3b19b6b-e1c3-499a-94eb-5db388e56f52	bridge	virbr0

```
[root@server1 ~]# nmcli con show ens160
```

```
connection.id: ens160
connection.uuid: ed943550-b2fd-4247-89ef-
connection.stable-id: --
connection.type: 802-3-ethernet
connection.interface-name: ens160
connection.autoconnect: yes
connection.autoconnect-priority: 0
connection.autoconnect-retries: -1 (default)
connection.multi-connect: 0 (default)
connection.auth-retries: -1
connection.timestamp: 1662325376
```

nmcli Command

- ❑ Remove an active connection

#nmcli con down [*connection name*]

- ❑ Attach a new or updated connection

#nmcli con up [*connection name*]

DNS Configuration



- ❑ The `/etc/resolv.conf` file is used to define the name servers that the system will use for DNS (domain name) resolution.
- ❑ Common directives:
 - `domain`: Specifies the DNS domain of the system.
 - `nameserver`: Specifies the IP address or hostname of a name server the system will use.

```
# /etc/resolv.conf
# Our domain
domain          vbrew.com
#
# We use vlager as central nameserver:
nameserver      191.72.1.1
```

Hostname Configuration

- ❑ View current hostname:

```
hostnamectl status
```

- ❑ Change hostname:

```
hostnamectl set-hostname myserver.local
```

/etc/services File



- ❑ Contains a list of **network ports** and the **services** that use them.
- ❑ Used by various networking programs to match service names with port numbers.
- ❑ When defining a **new service**, the system administrator must add a **service name** and **port number** pair to `/etc/services`.

❑ Port Ranges

- ❑ **Ports 0–1024:** Reserved ports (well-known, system-defined).
- ❑ **Ports >1024:** Can be assigned as needed by applications.

/etc/services: database of service names -> port/protocol
Format: service-name port/protocol [aliases] [# comment]

http	80/tcp	www	# World Wide Web HTTP
https	443/tcp		# HTTP over TLS/SSL
ssh	22/tcp		# Secure Shell
smtp	25/tcp	mail	# Simple Mail Transfer
domain	53/udp		# Domain Name Server
mysql	3306/tcp		# MySQL Database Service

Testing Network Connectivity



Tool	Purpose	Example
ping	Check if a host is reachable	ping google.com
traceroute	Show the path to a host	traceroute google.com
curl	Test HTTP/HTTPS request (get data)	curl https://example.com
wget	Download files from URLs	wget https://example.com
dig	Detailed DNS lookup	dig google.com
nslookup	Simple DNS query	nslookup google.com

Tip: Install `traceroute`, `dig`, and `nslookup` with `sudo yum install traceroute bind-utils`

Useful Network Troubleshooting Tools



Tool	Purpose	Example
<code>ss -tuln</code>	Show open TCP/UDP ports (replace <code>netstat</code>)	<code>ss -tuln</code>
<code>tcpdump</code>	Capture and analyze network traffic	<code>tcpdump -i eth0</code>
<code>ip -s link</code>	Show interface statistics (RX/TX)	<code>ip -s link</code>

Note:

- `netstat` is deprecated — use `ss` instead.
- `tcpdump` may need to be installed: `sudo yum install tcpdump`

Example

```
$ ss -tln
```

```
Netid  State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port
tcp    LISTEN  0        128     0.0.0.0:22          0.0.0.0:*
tcp    LISTEN  0        128     127.0.0.1:3306      0.0.0.0:*
udp    UNCONN  0         0       0.0.0.0:68          0.0.0.0:*
tcp    LISTEN  0        128     [::]:80            [::]:*
```

Column	Meaning
Netid	Protocol (e.g., TCP or UDP)
State	Connection state (e.g., LISTEN, UNCONN)
Local Address:Port	IP and port on the local system
Peer Address:Port	IP and port of the remote peer (* = any)

- This command is helpful for:
 - Checking which services are listening.
 - Troubleshooting open ports and firewall issues.

route Command (deprecated)

Overview:

- Used to **view**, **edit**, and **manage** the system's routing table.
- Allows admins to define **custom default routes**.
- Enables manual configuration of the **default gateway**.

Use Cases:

- Configure **static routes** to specific networks.
- Set or change the system's **default gateway**.

Note:

- `route` is **deprecated** in modern systems.
- Use `ip route` instead for new configurations.

route Command

□ Syntax:

```
[root@serverA /root]# route cmd type addy netmask mask gw gway dev dn
```

cmd

Either **add** or **del**, depending on whether you are adding or deleting a route. If you are deleting a route, the only other parameter you need is **addy**.

type

Either **-net** or **-host**, depending on whether **addy** represents a network address or a router address.

addy

The destination network to which you want to offer a route.

netmask mask

Sets the netmask of the **addy** address to **mask**.

gw gway

Sets the router address for **addy** to **gway**. Typically used for the default route.

dev dn

Sends all packets destined to **addy** through the network device **dn** as set by **ifconfig**.

route Command vs ip Command



Task	Legacy Command	Modern ip Command
View routing table	<code>route -n</code>	<code>ip route show</code>
Add route to network	<code>route add -net 192.56.76.0 netmask 255.255.255.0 eth0</code>	<code>ip route add 192.56.76.0/24 dev eth0</code>
Add default gateway	<code>route add default gw 192.168.1.254</code>	<code>ip route add default via 192.168.1.254</code>

Q&A

